American Computer Science League

2022-2023 ● Contest 1: Short Problems ● Junior Division

1. Computer Number Systems

Convert 2023_{10} to octal.

A. 2023
B. 3747
C. 4040
D. 4511
E. 4737

2. Computer Number Systems

What is the value of $657383_{16} - 654321_{16}$ in base 8?

A. 30242
B. 31142
C. 30152
D. 30142
E. 30132

3. Recursive Functions

Find $f(18)$ given:

$$f(x) = \begin{cases} f(x-3) + 1 & \text{if } x \geq 10 \\ f(x-2) - 3 & \text{if } 5 < x < 10 \\ x^2 & \text{if } x \leq 5 \end{cases}$$

A. 18
B. 19
C. 20
D. 21
E. 22

4. Recursive Functions

Find $f(12, 21)$ given:

$$f(x,y) = \begin{cases} f(x-1, y+2) - 3 & \text{if } x > y \\ f(y-1, x+2) + 1 & \text{if } x < y \\ x & \text{if } x = y \end{cases}$$

A. 13
B. 18
C. 20
D. 15
E. 12

5. What Does This Program Do? (Branching)

What is the output for this program given the following inputs:
a = 4, b = 5, c = 7

```
input a, b, c
if a^2 + b^2 == c^2 then
     ans = a * b / 2
else
     s = (a + b + c) / 2
     ans = sqrt(s * (s-a) * (s-b) * (s-c))
end if
output int(ans)
```

A. 96
B. 9
C. 10
D. 9.8
E. 14

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2022-2023 ● Contest 1: Short Problems ● Junior Division

1. 计算机数字系统

将2023_{10}转换为**八进制数。**

A. 2023
B. 3747
C. 4040
D. 4511
E. 4737

2. 计算机数字系统

$657383_{16} - 654321_{16}$的**值用八进制表示是多少？**

A. 30242
B. 31142
C. 30152
D. 30142
E. 30132

3. 递归函数

如果给定下列式子，请计算 $f(18)$

$$f(x) = \begin{cases} f(x-3) + 1 & \text{当 } x \geq 10 \text{ 时} \\ f(x-2) - 3 & \text{当 } 5 < x < 10 \text{ 时} \\ x^2 & \text{当 } x \leq 5 \text{ 时} \end{cases}$$

A. 18
B. 19
C. 20
D. 21
E. 22

4. 递归函数

如果给定下列式子，请计算 $f(12, 21)$

$$f(x,y) = \begin{cases} f(x-1, y+2) - 3 & \text{当 } x > y \text{ 时} \\ f(y-1, x+2) + 1 & \text{当 } x < y \text{ 时} \\ x & \text{当 } x = y \text{ 时} \end{cases}$$

A. 13
B. 18
C. 20
D. 15
E. 12

5. 此程序的作用是什么? （分支）

给定以下输入的值，该程序**会**输出什么：

a = 4, b = 5, c = 7

```
input a, b, c
if a^2 + b^2 == c^2 then
     ans = a * b / 2
else
     s = (a + b + c) / 2
     ans = sqrt(s * (s-a) * (s-b) * (s-c))
end if
output int(ans)
```

A. 96
B. 9
C. 10
D. 9.8
E. 14